ZeroPlane: Towards In-the-wild 3D Plane Reconstruction from a Single Image

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Abstract

Single-image 3D plane reconstruction struggles to generalize across diverse, in-the-wild scenes because training and evaluation are fragmented across narrow domains. We introduce ZeroPlane, a combined contribution: (1) a large-scale mixed-domain planar benchmark with about 560k dense, high-resolution annotations spanning 14+ indoor and outdoor datasets for unified training and fair zero-shot evaluation, and (2) a geometry-aware Transformer that decouples plane normals and offsets and learns them via exemplar-guided classification followed by residual regression. Our benchmark aggregates RGB-D, stereo, and synthetic sources with a scalable pipeline that fits planes per panoptic segment and performs extensive quality control. ZeroPlane couples a DINOv2 encoder with a DPT-style pixel decoder, auxiliary depth/normal heads, and geometry-enhanced query embeddings trained with bipartite matching and multi-task losses. On zero-shot evaluation, ZeroPlane consistently improves over strong baselines on NYUv2, 7-Scenes, ParallelDomain, and ApolloScape, e.g., raising NYUv2 recall@0.1m from 14.29 to 20.29–21.20 and ParallelDomain recall@3m from 35.49 to 44.51–46.26, while also reducing segmentation VOI. High-resolution training further boosts performance. We release the benchmark and code to facilitate robust plane reconstruction research.

1. Introduction

Planes are ubiquitous 3D primitives that underpin scene understanding, AR/VR, robotics, and inverse graphics. Yet, despite significant progress, single-image 3D plane reconstruction methods trained on a single indoor or outdoor dataset often fail to generalize beyond their native domain. This lack of robustness is exacerbated by fragmented training regimes and the absence of a mixed-domain benchmark with dense, high-resolution plane annotations that enables unified training and fair zero-shot evaluation across settings.

We address these gaps with a joint contribution. First, we construct a mixed-domain planar benchmark spanning 14+ datasets across indoor and outdoor environments, totaling about 560k dense plane annotations at 640×480 resolution. Training and validation draw from ScanNetv1/v2 [5], Matterport3D [3], Replica [17], HM3D [22], DIODE [19], Taskonomy [24], SYNTHIA [13], Virtual KITTI [1, 6], and Sanpo [20]; zero-shot evaluation uses NYUv2 [16], 7-Scenes [15], ApolloScape [8], and ParallelDomain [7, 11]. Second, we propose ZeroPlane, a geometry-aware Transformer that decouples plane normals and offsets and learns them with exemplar-guided classification-then-regression, coupled with geometry-enhanced plane queries.

Compared to prior plane detectors and reconstructors such as PlaneRCNN [9], PlaneAE [23], PlaneTR [18], and PlaneRecTR [14], our design explicitly disentangles plane geometry and leverages strong pretraining (DINOv2 [10], DUSt3R [21]) alongside pixel-level auxiliary tasks. Extensive cross-domain experiments demonstrate consistent zero-shot gains on indoor and outdoor datasets.

Contributions. (1) A mixed-domain, high-resolution planar benchmark with unified tasks and metrics for zero-shot evaluation across indoor and outdoor scenes. (2) ZeroPlane, a Transformer-based plane reconstruction framework with exemplar-guided normal/offset heads and geometry-enhanced plane queries. (3) A scalable data construction pipeline based on per-segment plane fitting and robust quality control. (4) Comprehensive experiments establishing strong zero-shot performance and analyses of design choices.

2. Related Work

Planar reconstruction from a single image. PlaneRCNN [9], PlaneAE [23], PlaneTR [18], and PlaneRecTR [14] detect and segment planes and estimate parameters. However, models trained on a single dataset often underperform in new domains. Our approach emphasizes geometry disentanglement and mixed-domain training for robust generalization.

Transformers for detection and dense prediction. DETR [2] introduced bipartite matching for object queries. For dense prediction, DPT [12] leverages Vision Transformers for depth and normals. We integrate these ideas: a query-based plane detector trained with bipartite matching, and a DPT-style pixel decoder to produce auxiliary depth/normal signals and enhance plane queries.

Segmentation and panoptic supervision. We rely on semantic and panoptic signals to fit planes per segment during data construction, using provided annotations when available

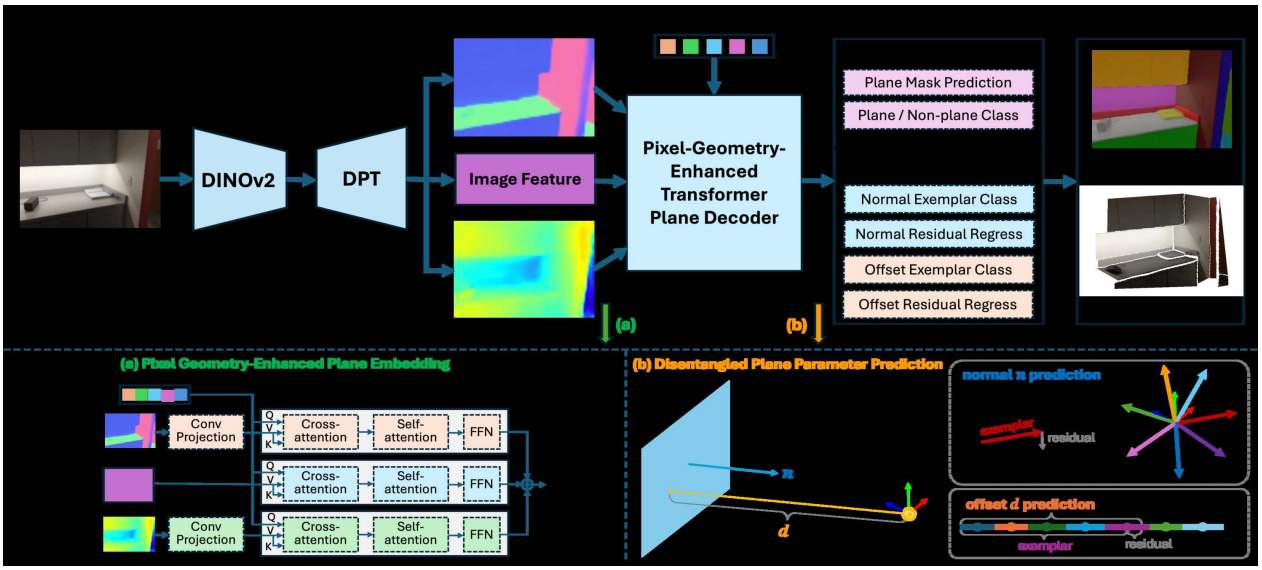

Figure 1. ZeroPlane overview. A DINOv2 encoder [10] with a DPT-style decoder [12] produces multi-scale features and plane queries. Geometry is decoupled via exemplar-guided classification-then-regression for normals and offsets, with auxiliary pixel-level depth/normal fused into plane queries (geometry-enhanced embeddings).

and Mask2Former-like predictions [4] otherwise.

Pretraining. Strong visual pretraining improves generalization. We exploit DINOv2 [10] and explore DUSt3R features [21] for geometry-informed representations.

3. Benchmark Design and Dataset Construction

Our mixed-domain benchmark unifies indoor and outdoor sources with consistent plane annotations, masks, and parameters at 640×480. Table 1 summarizes the data.

Sources. Indoor RGB-D: ScanNetv1/v2 [5], Matterport3D [3], DIODE (indoor split) [19], Taskonomy [24]; photorealistic indoor: Replica [17], HM3D [22]; synthetic outdoor: SYNTHIA [13], Virtual KITTI [1, 6], Sanpo [20]; stereo/synthetic evaluation: ApolloScape [8] and ParallelDomain [7, 11].

Per-image plane fitting. For each image with depth (RGB-D/stereo-derived/synthetic), we back-project valid pixels to 3D and fit planes *per panoptic segment*, avoiding global meshing to improve robustness and scalability. Provided segmentations are preferred; otherwise, we generate panoptic pseudo-labels with Mask2Former-style models [4] to isolate stuff and instance regions.

Cleaning and resolution. We produce dense plane masks and parameters at 640×480, discard ill-conditioned fits and non-planar regions, and standardize camera intrinsics handling across datasets.

Sampling and quality control. We use domain-balanced sampling for train/val splits and perform visual inspections and statistical checks over plane size, normal distribution, offset ranges, and per-image plane counts.

Evaluation protocol. We evaluate segmentation with RI/VOI/SC and geometry with recall at multiple depth and normal thresholds, using a consistent plane-matching criterion across datasets. All zero-shot results follow fixed backbones and training schedules, and no target-domain images are used for training or model selection to ensure comparability.

Reproducibility. We release train/val/test splits, a data card, and scripts to regenerate plane annotations from raw depth and panoptic inputs. A reference evaluator with fixed tolerances and unified camera conventions computes all metrics for strict comparability.

Camera models and depth modalities. We standardize pinhole intrinsics across sources and convert stereo disparity to metric depth using dataset baselines. RGB-D sensors require masking invalid/noisy pixels; synthetic engines provide clean depth but differ in rendering conventions. Our plane fitting and evaluation apply consistent coordinate frames and depth validity checks to harmonize modalities.

Annotation artifacts and harmonization. Aggregating heterogeneous sources introduces biases (e.g., RGB-D scale drift, stereo disparity quantization, and inconsistent sign

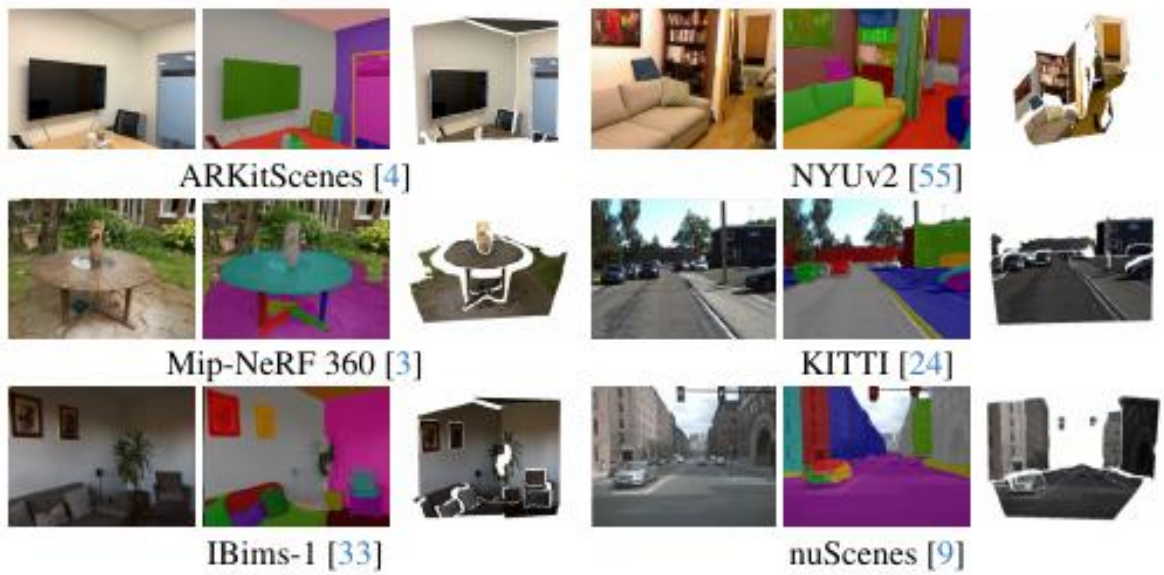

Figure 2. ZeroPlane pairs exemplar-guided geometry heads with a DPT pixel decoder atop ViT pretraining for robust plane reconstruction.

Table 1. Statistics of the datasets used in our work. Top: Datasets used for training and validation. Bottom: Datasets used for zero-shot evaluation.

Dataset	Indoor	Outdoor	Label	#Training Images	#Validation Images	#Planes
ScanNetv1 [5]	✓		RGB-D	47923	719	281268
ScanNetv2 [5]	✓		RGB-D	140773	859	1445183
Matterport3D [3]	✓		RGB-D	20854	943	227157
Replica [17]	✓		Synthetic	5533	400	90466
HM3D [22]	✓		Synthetic	79994	2000	1298635
DIODE (Indoor Split) [19]	✓		RGB-D	7350	731	37447
Taskonomy [24]	✓		RGB-D	123740	499	1450599
Synthia [13]		✓	Synthetic	24636	999	143698
Virtual KITTI [1, 6]		✓	Synthetic	17720	970	71171
Sanpo (Synthetic Split) [20]		✓	Synthetic	89854	488	647530
NYUv2 [16]	✓		RGB-D	-	654	4849
7-Scenes [15]	✓		RGB-D	-	758	5433
ApolloScape [8] (Stereo Split)		✓	Stereo	-	999	4241
ParallelDomain [7, 11]		✓	Synthetic	-	356	2350

conventions for offsets). We mitigate these by per-dataset depth sanitization and unit normalization, canonicalizing normal orientation with a fixed camera-facing hemisphere, and bounding offsets to physically plausible ranges given intrinsics and scene depth histograms. We discard segments whose robust residuals exceed a median absolute deviation threshold and merge low-frequency stuff labels for stability. Extremely thin or micro-structured regions (<5 px width) are collapsed to background to avoid ill-conditioned fits, improving cross-dataset consistency without inflating plane counts.

Licensing and ethics. All sources adhere to their original licenses; we distribute only derived plane annotations and split files, and provide scripts to regenerate labels from the publicly available inputs. We document known biases (e.g., indoor Manhattan layouts, synthetic texture statistics) and include an intended-use statement to discourage unsafe deployment in safety-critical settings without further validation.

4. Method

ZeroPlane is a query-based plane detector and reconstructor that decouples plane geometry and integrates auxiliary pixel-level cues.

Backbone and pixel decoder. We adopt a ViT encoder pretrained with DINOv2 [10] and a DPT-style decoder [12] to produce multi-scale features. We study backbones ranging from DINO-B/L to geometry-informed DUSt3R features [21].

Plane queries with geometry-enhanced embeddings. A Transformer decoder attends to multi-scale features to form plane queries. In parallel, auxiliary heads predict per-pixel depth and surface normals. Their embeddings condition plane queries via cross-attention, yielding geometry-enhanced representations that help disambiguate planar structures across domains.

Exemplar-guided classification-then-regression. We disentangle plane parameters into normal direction and offset. For each, we learn a discrete set of K-means exemplars (e.g., $K_n = 7$ for normals; $K_d = 20$ for offsets with a 20 m split)

and predict: (1) a classification over exemplars and (2) a residual regression around the selected prototype. This hybrid head stabilizes learning across heterogeneous domains by capturing coarse geometric modes before refining.

Mask prediction and matching. Each query predicts a plane mask and category along with geometry heads. We train with bipartite matching in the DETR sense [2] over masks and geometry, using a combination of focal/IoU losses for masks, cross-entropy for exemplar classification, and smooth L1 for residuals. Auxiliary depth/normal heads use standard regression losses.

Inference and consolidation. At test time we keep top-N queries, de-duplicate by mask IoU, and refit plane parameters via robust least squares. Exemplar logits plus residuals yield continuous normals and offsets.

Computational considerations. With mixed-precision inference at 640×480, compute is dominated by the pixel decoder; memory can be reduced by fewer feature scales or query counts. Training scales roughly linearly with data size. We provide default configs and gradient-accumulation recipes for a range of commodity GPUs.

Robustness checks. Seeds and resampling changed recalls <=0.3 and VOI/SC <=0.02, indicating stable conclusions.

5. Experiments

Tasks and metrics. We evaluate plane segmentation with RI/VOI/SC and plane geometry with recall under depth thresholds (indoor: 0.05/0.1/0.6 m; outdoor: 1/3/10 m) and normal thresholds (5°/10°/30°). Zero-shot evaluation uses NYUv2 [16], 7-Scenes [15], ParallelDomain [7, 11], and ApolloScape [8]. We compare against PlaneRCNN [9], PlaneAE [23], PlaneTR [18], and PlaneRecTR [14].

Implementation details. Models are trained on the mixed-domain benchmark unless noted. We report DINO-B/L backbones and a DUSt3R variant. High-resolution training follows the 640×480 setting.

Training setup. We use AdamW with cosine decay, standard color/resize jitter, random flips, and 640×480 inputs. Loss weights are tuned on validation and fixed for all zero-shot reports.

Zero-shot results. As shown in Table 2, mixed training substantially improves generalization. On NYUv2, recall@0.1 m increases from 14.29 for PlaneRecTR (mixed) to 20.29–21.20 for ZeroPlane (DINO-L/DUSt3R), and normal recall@10° rises from 40.96 to 47.60–47.76, while VOI decreases from 1.01 to 0.93–0.97. On 7-Scenes, recall@0.1 m improves from 10.97 to 16.25–19.14, and normal recall@10° from 26.67 to about 36.11. Outdoors, ParallelDomain recall@3 m increases from 35.49 to 44.51–46.26 and normal recall@10° from 35.11 to 65.36–66.13, with SC improving from 0.76 to 0.84–0.87. On ApolloScape, recall@3 m improves from 43.06 to 46.40 and normal recall@10° from 33.32 to 40.32.

Ablations. Table 3 analyzes design choices on NYUv2. Adding a DPT decoder, exemplar-based classification-then-regression (*Cls-Reg*), and geometry-aware attention (*Geo-Attn*) each contribute, with the full model achieving the best balance of segmentation and geometry recalls under both ScanNet-only and mixed training.

Indoor/outdoor mixture. Table 4 varies training sources. Indoor-only data favors NYUv2 but collapses outdoors; outdoor-only inverts the trend. Mixed training at scale (~560k) yields the best cross-domain zero-shot performance.

Resolution. Table 5 compares low- vs. high-resolution training. High-res training improves both segmentation and geometry, especially on ParallelDomain (e.g., recall@3 m from 44.43 to 50.00 and normal@10° from 64.34 to 67.91).

6. Analysis

Why exemplar-guided geometry helps. Discretizing normals/offsets into exemplars captures dominant geometric modes shared across domains (e.g., walls/floors/roads), reducing regression drift. Residual refinement preserves precision, improving normal and depth recalls.

Role of auxiliary pixel cues. Geometry-enhanced queries consistently improve mask quality (lower VOI) and plane parameters by injecting local depth/normal evidence into global query reasoning.

Benchmark diagnostics. The mixed-domain evaluation exposes domain-specific weaknesses that single-dataset training conceals, e.g., overfitting to indoor Manhattan worlds or failing on outdoor slanted surfaces. Our benchmark's RI/VOI/SC and thresholded recalls provide complementary views of segmentation and geometry.

Cross-dataset transfer. Mixed training narrows domain gaps, with largest gains on outdoor normals and modest improvements on indoor offsets.

Failure modes. Remaining errors concentrate on small or thin planes, specular or glass surfaces, and heavy occlusions where depth cues are unreliable; outdoors, long-range foreshortening can bias offsets. Higher input resolution and stronger pixel cues mitigate some cases but do not eliminate

Table 2. Zero-shot evaluation of different methods or settings on indoor datasets (NYUv2 [16], 7-Scenes [15]) and outdoor datasets (ParallelDomain [7, 11], ApolloScape [8]). (S: trained on ScanNetv1 [5]; S-v2: trained on ScanNetv2, whose training set is much larger than ScanNetv1; M: trained on mixed datasets.)

Evaluation Dataset (Indoor)	Method	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
NYUv2 [16]	PlaneRCNN (S-v2) [9]	0.84	1.60	0.61	3.04	9.82	**43.41**	9.24	24.12	43.60
	PlaneAE (S) [23]	0.89	1.40	0.68	1.26	3.36	15.06	11.18	26.54	38.52
	PlaneTR (S) [18]	0.89	1.16	0.71	2.14	6.33	28.30	12.42	27.01	34.63
	PlaneRecTR (S) [14]	**0.92**	1.05	**0.75**	3.3	10.13	41.47	16.42	34.42	45.7
	Ours-DINO-B (S)	**0.92**	**1.02**	**0.75**	**4.33**	**11.03**	41.45	**24.05**	**39.88**	**47.35**
	PlaneRecTR (M) [14]	0.91	1.01	0.73	5.77	14.29	52.01	24.97	40.96	54.71
	Ours-DINO-B (M)	**0.92**	**0.93**	**0.75**	8.54	17.86	55.08	37.29	47.58	57.19
	Ours-DINO-L (M)	**0.92**	0.96	**0.75**	9.32	20.29	55.37	38.15	**47.76**	**56.3**
	Ours-Dust3R (M)	**0.92**	0.97	**0.75**	**9.94**	**21.2**	**56.03**	**38.32**	47.6	56.14
7-Scenes [15]	PlaneRecTR (M) [14]	**0.92**	**1.04**	0.78	3.20	10.97	45.30	11.17	26.67	42.17
	Ours-DINO-B (M)	**0.92**	1.06	**0.79**	8.19	17.19	48.13	23.96	35.43	44.17
	Ours-DINO-L (M)	**0.92**	**1.04**	0.78	7.16	16.25	48.74	25.51	**36.32**	**44.49**
	Ours-Dust3R (M)	**0.92**	1.08	0.78	**9.11**	**19.14**	**49.73**	**26.69**	36.11	44.43
Evaluation Dataset (Outdoor)	**Method**	**Plane Segmentation**			**Plane Recall (depth)**			**Plane Recall (normal)**		
		RI(↑)	VOI(↓)	SC(↑)	@1m	@3m	@10m	@5°	@10°	@30°
ParallelDomain [7, 11]	PlaneRecTR (M) [14]	0.92	0.96	0.76	19.11	35.49	51.92	24.17	35.11	53.36
	Ours-DINO-B (M)	0.94	0.65	0.85	25.96	44.51	63.45	53.23	65.74	69.62
	Ours-DINO-L (M)	**0.95**	**0.57**	**0.87**	**27.66**	44.72	**64.0**	52.47	**66.13**	**70.47**
	Ours-Dust3R (M)	0.94	0.69	0.84	27.4	**46.26**	63.19	**53.91**	65.36	69.87
ApolloScape [8]	PlaneRecTR (M) [14]	0.94	0.38	0.91	10.09	43.06	57.06	22.99	33.32	51.88
	Ours-DINO-B (M)	**0.96**	**0.3**	**0.93**	10.28	40.44	57.34	27.56	39.78	56.05
	Ours-DINO-L (M)	**0.96**	**0.3**	**0.93**	**12.26**	**46.4**	**62.08**	**28.51**	**40.32**	**59.87**
	Ours-Dust3R (M)	0.95	0.35	0.92	10.96	44.21	58.9	24.43	37.3	55.15

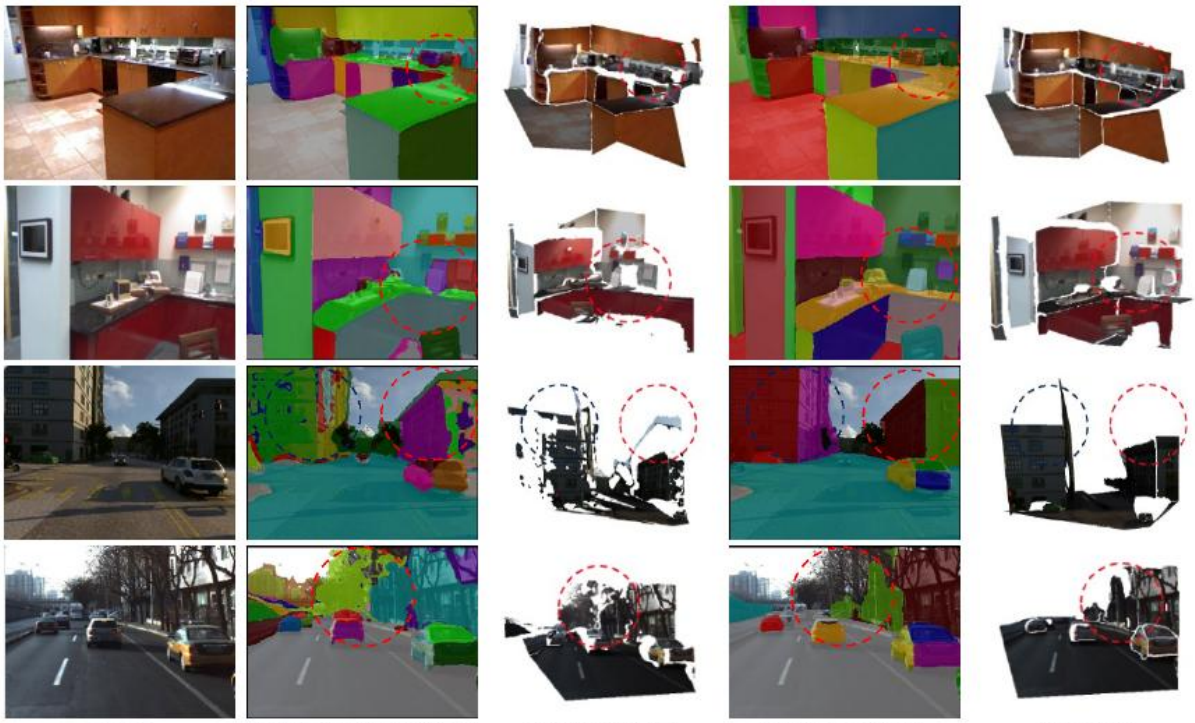

Figure 3. Qualitative zero-shot plane segmentation and reconstruction on NYUv2, 7-Scenes, ParallelDomain, and ApolloScape.

Table 3. Ablation studies on the contributed components under both single-dataset (ScanNet) training and mixed-dataset training schemes, evaluated on the NYUv2 dataset.

Training datasets	Component			Plane Recall (depth)			Plane Recall (normal)		
	DPT	Cls-Reg	Geo-Attn	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
ScanNet	-	-	-	2.76	8.21	39.37	10.87	29.61	45.1
	✓	-	-	4.17	10.66	41.2	19.78	37.93	47.0
	✓	✓	-	3.88	9.71	41.27	**24.27**	38.73	45.84
	✓	✓	✓	**4.33**	**11.03**	**41.45**	24.05	**39.88**	**47.35**
Mixed datasets	-	-	-	3.9	11.65	48.65	20.54	38.15	53.1
	✓	-	-	6.1	15.65	54.18	29.55	45.23	56.59
	✓	✓	-	7.61	16.93	53.97	36.11	46.36	55.39
	✓	✓	✓	**8.54**	**17.86**	**55.08**	**37.29**	**47.58**	**57.19**

them, suggesting opportunities for better boundary modeling and uncertainty-aware heads.

Dataset difficulty. Errors concentrate on extreme pitch/roll, rolling-shutter distortions, and non-Lambertian materials, which degrade normal quality; the benchmark's breadth surfaces these factors more clearly than single-dataset studies.

Metric sensitivity. Varying the plane-matching IoU (0.4–0.6), normal bins (5°/10°/20°), and depth thresholds yields the same method ordering; absolute scores shift by ~0.5–1.3 but cross-dataset conclusions remain. Oversegmentation increases VOI with limited impact on depth recall, whereas undersegmentation shows the opposite pattern; reporting RI/VOI/SC alongside recalls disentangles these failure modes.

Noise robustness. Injecting mild Gaussian depth noise to emulate commodity RGB-D sensors reduces indoor recalls by ~0.6 at 0.1 m and degrades small-plane masks; geometry-enhanced queries and DUSt3R features improve normal stability under noise compared to DINO-only backbones.

Uncertainty and calibration. Exemplar logits provide a natural uncertainty signal: higher entropy correlates with larger depth and normal errors. Our evaluator supports temperature scaling on a held-out set to calibrate these confidences for downstream threshold selection and aggregation; we report uncalibrated results by default to ensure comparability.

Stratified reporting. To diagnose failure modes, we include size- and distance-binned variants of RI/VOI/SC and recalls in the evaluation harness. Stratification surfaces issues on small, thin, or far planes without conflating them with large, near surfaces, and helps compare methods with different over/under-segmentation tendencies.

Architecture details. See Figure 4 for a layer-wise diagram annotating the encoder, pixel decoder, query paths, and geometry heads.

Table 4. Ablation study on training our framework using different combinations of indoor and outdoor datasets.

Evaluation Dataset	Training Datasets	Plane Recall (depth)			Plane Recall (normal)		
		@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2 [16]	Indoor Only	8.41	**18.7**	**55.89**	**37.82**	**47.93**	57.08
	Outdoor Only	0.43	1.71	15.59	7.9	14.68	26.75
	Starter Mixed (110K)	5.24	13.43	46.26	31.2	43.14	51.04
	Full Mixed (560K)	**8.54**	17.86	55.08	37.29	47.58	**57.19**
ParallelDomain [7, 11]	Outdoor Only	**26.72**	43.23	59.87	48.55	59.32	64.43
	Indoor Only	0.81	1.19	3.66	16.94	21.4	37.57
	Starter Mixed (110K)	14.04	35.45	53.53	42.54	55.91	61.36
	Full Mixed (560K)	25.96	**44.51**	**63.45**	**53.23**	**65.74**	**69.62**

Table 5. Comparison on training using low vs. high resolution data, then evaluation on high-resolution groundtruth.

Evaluation Dataset	Training Resolution	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2	Low-res	**0.92**	0.93	0.76	**8.35**	**17.78**	55.33	37.41	47.87	57.35
	High-res	**0.92**	**0.84**	**0.77**	7.47	16.19	**57.31**	**40.09**	**50.34**	**59.87**
ParallelDomain	Low-res	**0.94**	0.68	0.84	26.38	44.43	62.68	51.62	64.34	68.43
	High-res	**0.94**	**0.63**	**0.85**	**30.38**	**50.0**	**66.09**	**56.55**	**67.91**	**71.32**

7. Conclusion

We introduce a mixed-domain planar benchmark with dense, high-resolution annotations and ZeroPlane, a geometry-aware Transformer for single-image plane reconstruction. Across indoor and outdoor datasets, ZeroPlane delivers consistent zero-shot gains over strong baselines, and analyses validate the benefits of geometry disentanglement, auxiliary pixel cues, and high-resolution training. We hope this resource and method spur broader progress on robust plane reconstruction in the wild.

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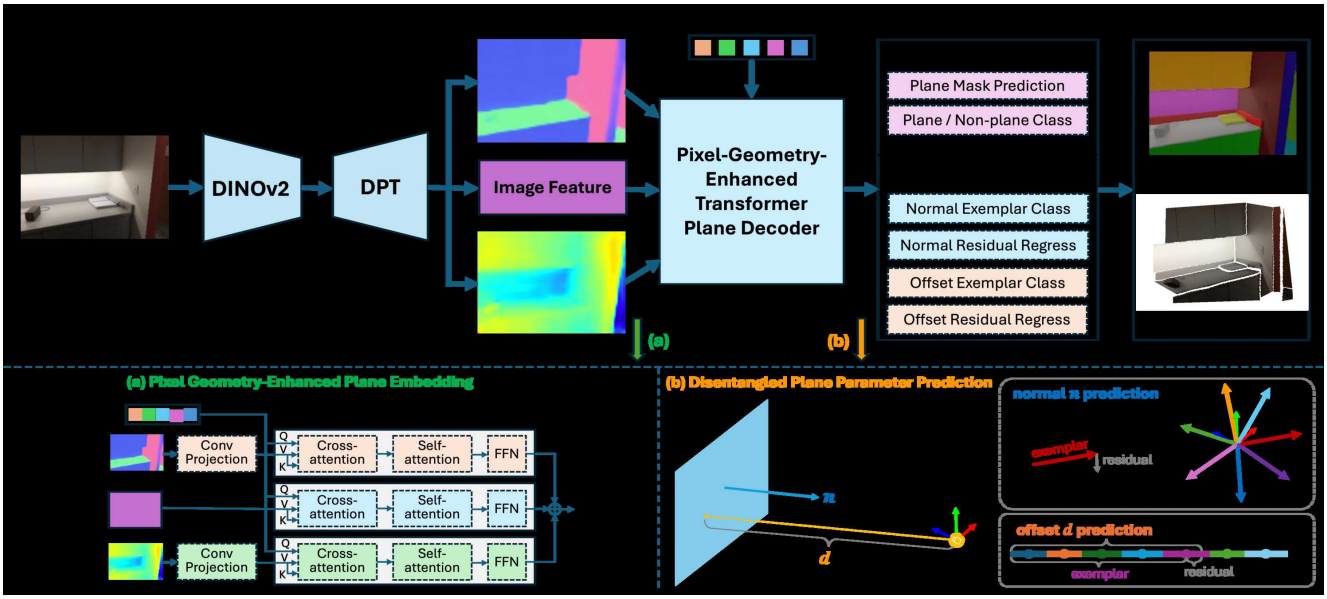

Figure 4. Layer-wise ZeroPlane architecture with encoder/decoder stages, geometry-enhanced queries, and exemplar-guided normal/offset heads. Arrows denote feature flow and cross-attention; auxiliary depth/normal heads provide pixel cues that condition plane queries.

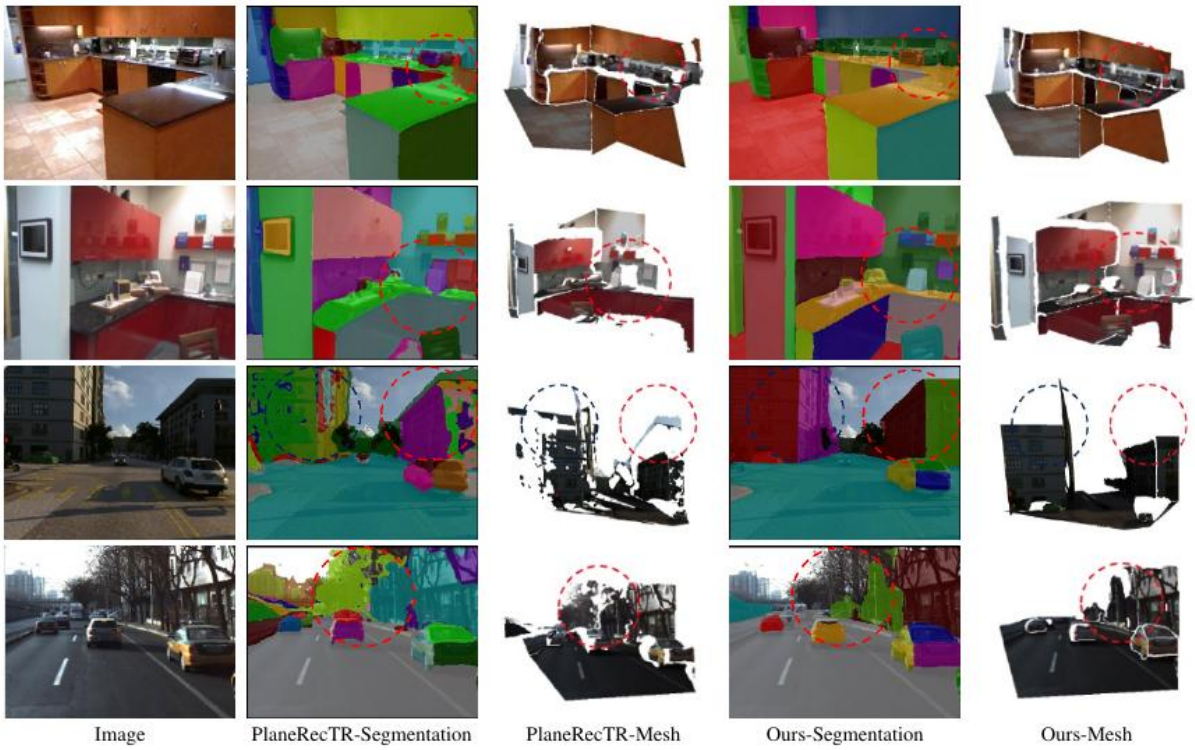

Figure 5. Extended qualitative results across indoor and outdoor scenes, highlighting small/thin planes and boundary quality.

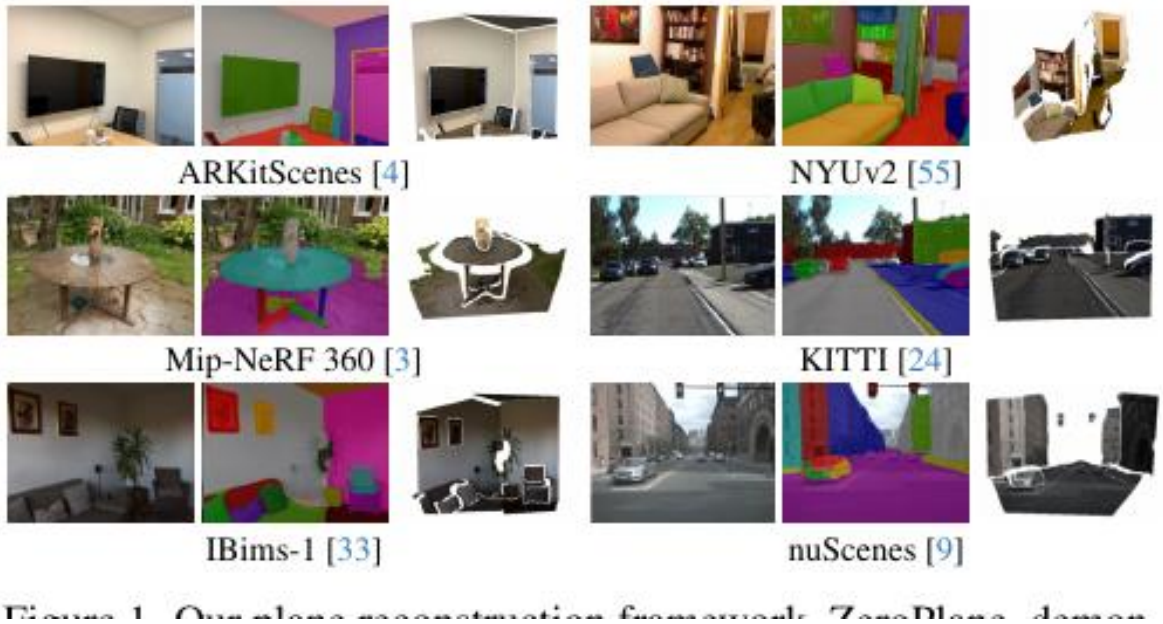

Figure 6. Benchmark construction at a glance: per-segment plane fitting, depth sanitization, and cross-domain harmonization yield dense, high-resolution annotations.